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Game Design Document
AimLock

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Aim Trainer Game Proposal

Overall Game Idea

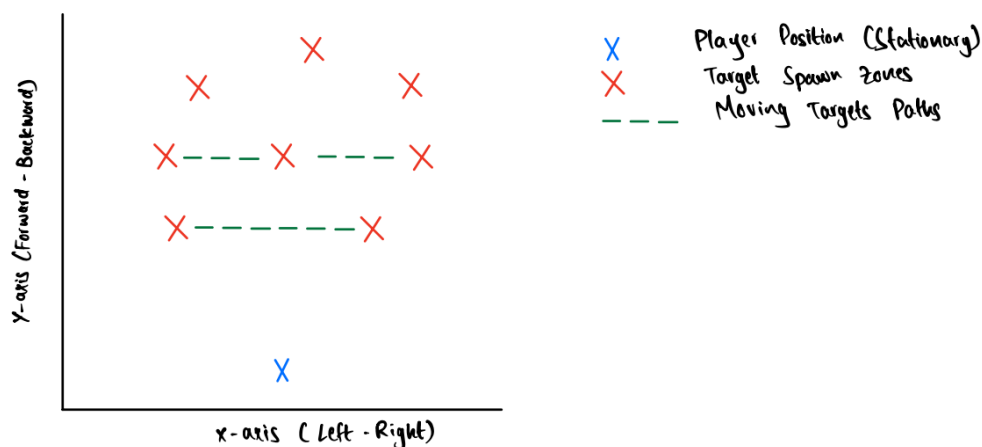
This project is a simplified aim trainer game inspired by Aimlabs. The game is designed to help players improve their aiming skills through target shooting. It will feature mouse-only controls with customizable sensitivity, crosshairs, environments, and target settings. Unlike FPS games, there will be no character movement (WASD), only aiming and shooting. The game will track accuracy and reaction time while allowing users to modify their training experience.

The game will include various training modes, such as:

- Static Targets – Targets appear at fixed positions for accuracy training.
- Moving Targets – Targets move across the screen to enhance tracking skills.
- Reaction Training – Targets appear randomly, challenging reaction time. (Stretch Goal)
- Flick Training – Targets pop up briefly, requiring fast flick shots. (Stretch Goal)
- Precision Mode – Targets shrink over time, emphasizing pixel-perfect accuracy. (Stretch Goal)

The game aims to be a lightweight, accessible, and highly customizable training tool for both casual and competitive gamers. With an intuitive UI and a quick restart feature, users can instantly reset sessions and switch between different training modes. This ensures they maximize their practice time without unnecessary interruptions.

Game Level Layout



Player Mechanics

Controls:

- **Mouse Movement:** Controls the aiming reticle (crosshair).
- **Left Click:** Fires a shot at the target.
- **Escape Key:** Opens the settings menu (pause, customization options).
- **Right Click:** Can be assigned to a secondary interaction like zooming.

Abilities:

- **Adjust Sensitivity:** Fine-tune mouse sensitivity to replicate FPS game settings.
- **Crosshair Customization:** Modify size, color, thickness, shape, and center gap.
- **Target Interaction:** Shoot at static or moving targets to track accuracy.
- **Performance Tracking:** Real-time stats display accuracy and shot percentage.

Animation State

Since this is a stationary aim trainer with no movement animations, the only animations involved would be for UI interactions and shooting effects:

- **Idle:** Default aiming state.
- **Shooting:** When the left-click is pressed, a firing effect may be triggered.
- **Hit Detection:** Targets disappear upon being hit.
- **Menu Interaction:** UI transitions when opening settings or modifying customization.

Target Type 1: Static Target

- **Abilities & Behavior:**
 - Remains in a fixed position.
 - Disappears upon being shot.
 - Spawns at pre-determined or randomized positions.
- **State Machine:**
 - Idle (waiting to be hit).
 - Hit detection (if shot, register hit).
 - Despawn (disappears after a set time or upon being hit).
 - Respawn (new target appears in a different location).
- **Animations:**
 - Target disappears when hit.

Target Type 2: Moving Target

- **Abilities & Behavior:**
 - Moves along a **predefined path**.
 - Can change speed based on difficulty settings.

- Disappears after being hit or after a set time.
- **State Machine:**
 - Spawn (target appears at a spawn point).
 - Move (target follows a defined path).
 - Hit detection (if shot, register hit).
 - Despawn (target disappears after hit or after a time limit).
 - Respawn (new target appears).
- **Animations:**
 - Target disappears upon hit.
 - Smooth movement animations.

Other Game Entities

Triggers

- **Target Spawning Triggers:** Determines when and where targets appear.
- **Scoring & Performance Tracking:** Tracks accuracy, reaction time, and streak count.
- **Mode Selection:** Allows players to switch between different training modes.

Other Relevant Features

- **Customization UI:** Allows seamless adjustments to settings without restarting the session.
- **Sound & Visual Feedback:** Improves the immersive experience by giving immediate response upon hitting targets.